Hypothesis Testing $\quad p = P(\text{heads})$

$H_0: p = 0.5$

$H_1: p \neq 0.5$

What is my tolerance for type I error

Type I error - reject H_0 even though H_0 is true

Choose $\alpha = P(\text{Type I error})$

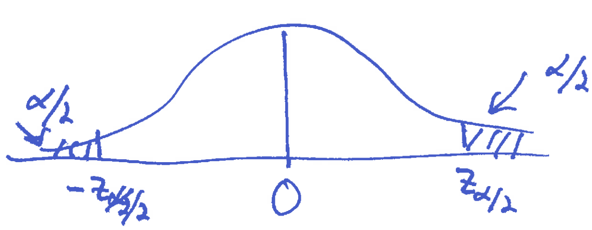
Based on that, construct rejection region, i.e.
for which values of test stat. would I reject H_0?

Here: Could use $T = \dfrac{\hat{p} - 0.5}{\sqrt{\frac{(0.5)(0.5)}{n}}}$

n ← # of tosses

1. Says something about the H_0 vs. H_1
2. know dist. of T under H_0 (at least approx.)
(namely, $T \sim N(0,1)$)

So, should reject H_0 if $T > z_{\alpha/2}$ or $T < -z_{\alpha/2}$



If $p = 0.51$, what is the prob. of rejecting H_0?
Answer will depend on n

Theory Lecture: Simple Linear Regression

This lecture provides an overview of the most basic regression model, the **simple linear regression model**. There is a single **covariate/predcitor** x used to predict the **response** Y. This model will serve as a building block for the more complex (and more useful) models we will present afterwards.

The simple linear regression model is typically written using notation

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, 2, \ldots, n$$

where β_0 is the **intercept** and β_1 is the **slope** for the regression line.

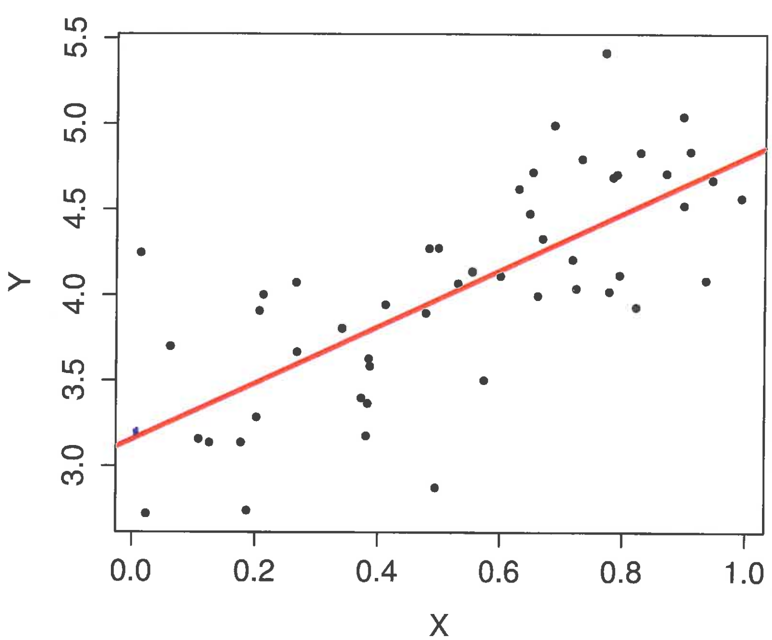


Figure 1: Synthetic scatter plot with $n = 50$ and a fitted regression line, $Y = 3.15 + 1.65x$.

The ϵ_i are random variables capturing the **irreducible error**, the scatter around the regression line.

The exact assumptions placed on the ϵ_i will vary depending on the situation, but the base assumptions that are almost always associated with simple linear regression are as follows:

1. $E(\epsilon_i) = 0 \quad$ for all i
2. $V(\epsilon_i) = \sigma^2 \quad$ for all i
3. The ϵ_i are uncorrelated, i.e., $\text{Cov}(\epsilon_i, \epsilon_j) = 0$ for all $i \neq j$.

Additional assumptions, **if justified**, lead to stronger results. In particular, one often assumes that the ϵ_i are normally distributed.

Exercise: Discuss the practical implications of these assumptions.

Assuming $E(\epsilon_i) = 0$ implies an "even distribution" of points around the line

$V(\epsilon_i) = \sigma^2$ implies similar "scatter" around the line

Assuming that $\text{Cov}(\epsilon_i, \epsilon_j) = 0$ implies a lack of linear relationship between the errors

Example: This model is sometimes written as

$$Y = \alpha + \beta x + \epsilon.$$

This notation is the origin for references to the "beta" for a stock.

Figure 2: Screen shot from Google Finance, taken September 3, 2016.

In particular, the "beta" is found my performing a simple linear regression where the response is the excess returns for that equity, and the predictor is the excess returns for the market (as measured by some standard index such as S & P 500).

Figure 3 shows an example of the use of simple linear regression with the objective of calculating the "beta" for Google (GOOG), using daily market from 2014.

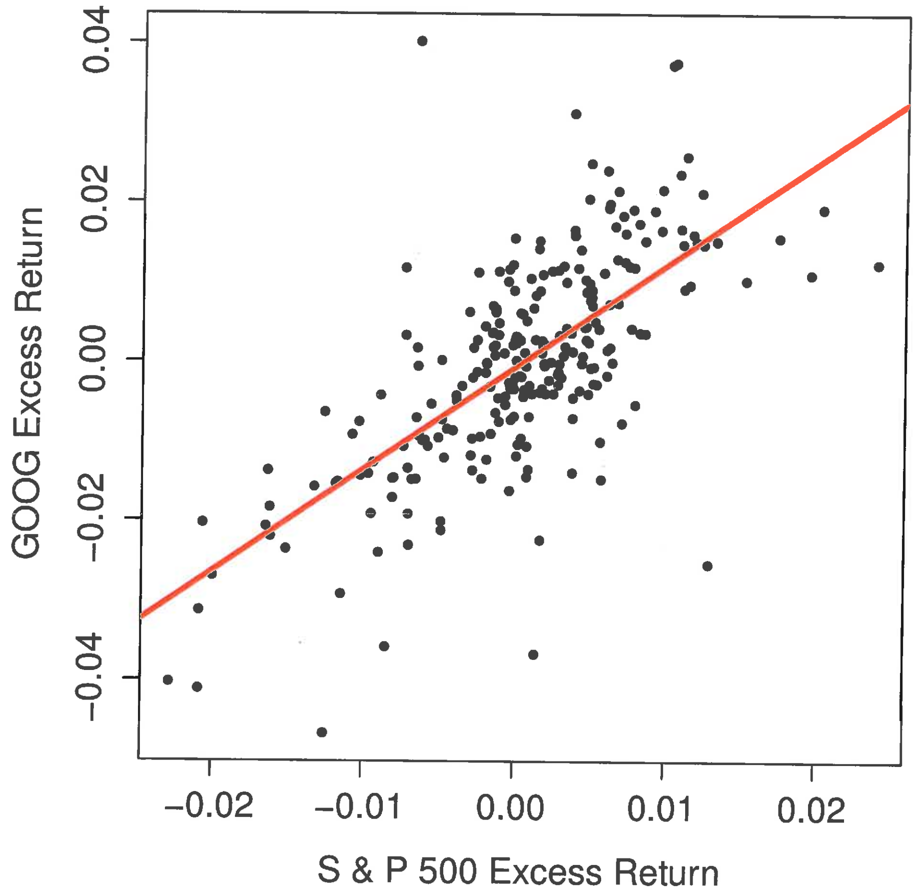


Figure 3: Daily excess returns for GOOG versus daily excess returns for S & P 500 for 2014.

Estimation via a Sample

In practice, we rely on a sample of n pairs (x_i, y_i) for $i = 1, 2, \ldots, n$, assumed to be drawn from a larger population. These will be used to estimate β_0, β_1 and σ^2. The standard notation for these estimates will be $\widehat{\beta}_0$, $\widehat{\beta}_1$, and $\widehat{\sigma}^2$.

The sample is naturally depicted in a **scatter plot**, with example shown in Figure 4.

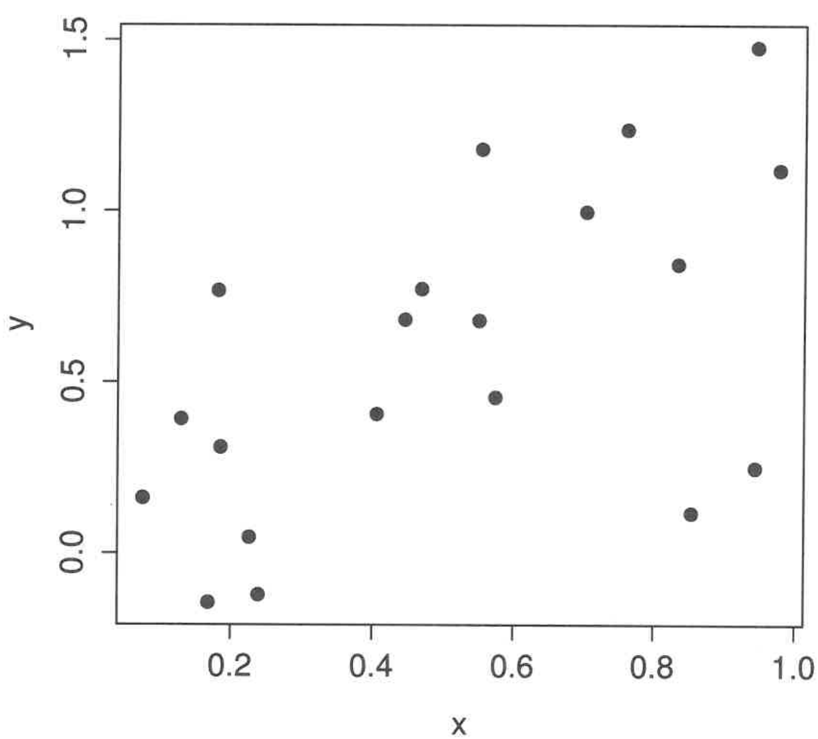


Figure 4: A simple scatter plot, with $n = 20$.

Some additional notation we should establish immediately: For any estimated regression line, the **fitted values** are

$$\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i, \quad i = 1, 2, \ldots, n$$

and the **residuals** are

$$\widehat{\epsilon}_i = y_i - \widehat{y}_i, \quad i = 1, 2, \ldots, n.$$

$$\approx \epsilon_i = Y_i - (\beta_0 + \beta_1 X_i)$$

Exercise: Use the scatter plot below, with a fit regression line superimposed, to demonstrate the fitted values and the residuals.

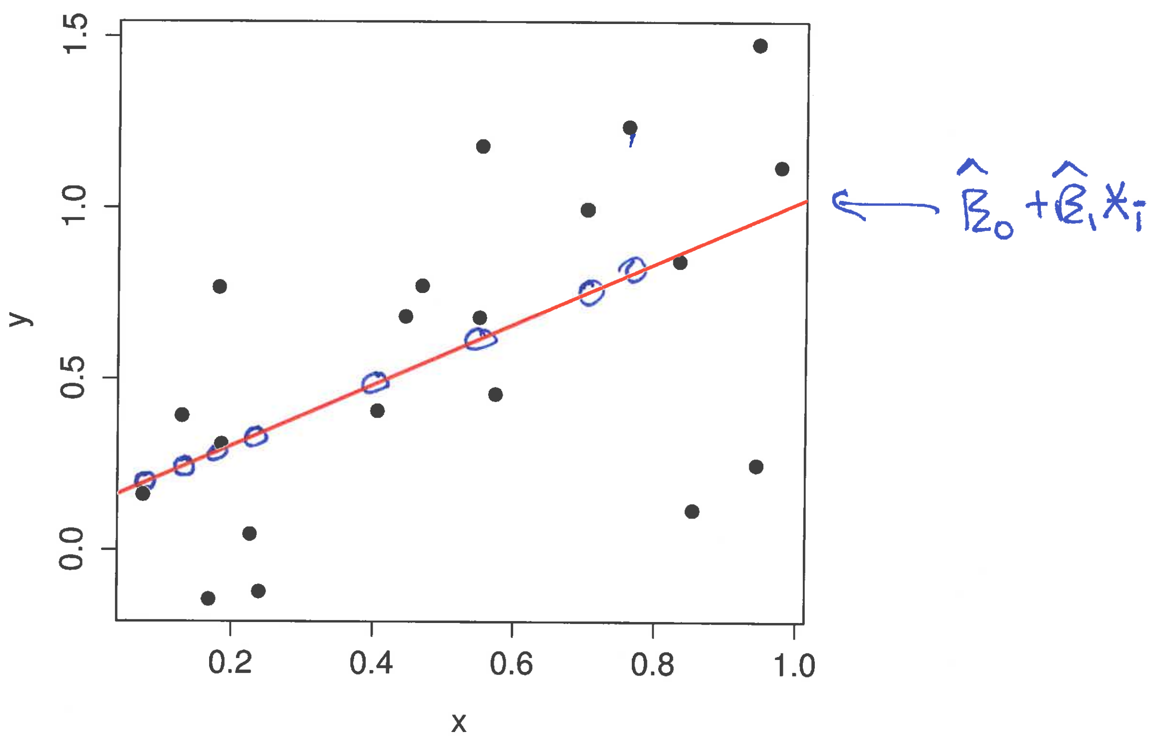


Figure 5: A simple scatter plot, with $n = 20$, with a regression line fit shown.

The Sample Covariance and Correlation

⋆ **Exercise:** Contrast covariance and correlation.

Recall that the covariance (and a related quantity, correlation) are measures of the **linear** association between two random variables X and Y:

$$\mathrm{Cov}(X,Y) = \mathrm{E}[(X - \mathrm{E}(X))(Y - \mathrm{E}(Y))]$$

and

$$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

Not surprisingly, sample versions of these quantities are relevant to simple linear regression.

First, the **sample covariance** between observations of n pairs (x_i, y_i):

$$s_{xy} = \left(\frac{1}{n-1}\right) \sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y})$$

Note the similarity between this and the usual sample variance:

$$s_x^2 = \left(\frac{1}{n-1}\right) \sum_{i=1}^{n} (x_i - \overline{x})^2$$

Then, the **sample correlation** is

$$r_{xy} = \frac{s_{xy}}{s_x s_y}.$$

This quantity is typically just denoted with r. Figure 6 shows some examples of scatter plots, and the value of r for each. It is useful for building some intuition as to what different values of r mean.

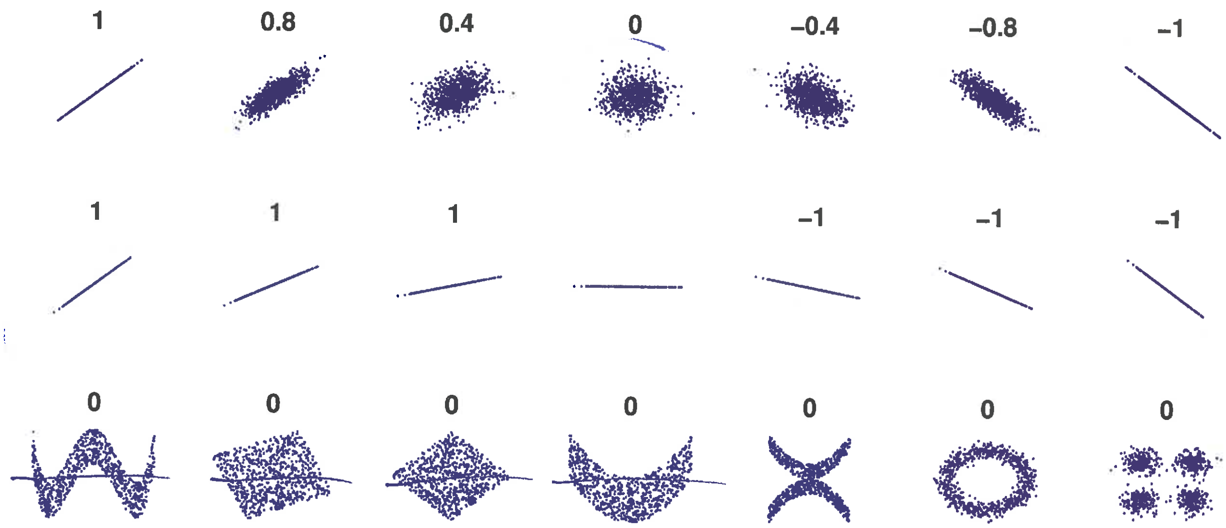
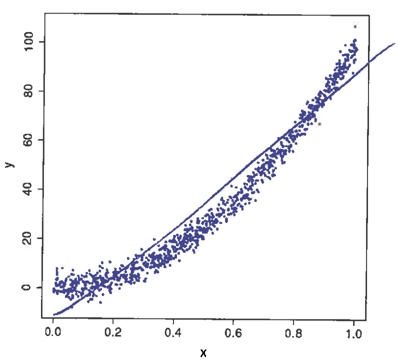


Figure 6: Examples of scatter plots, and corresponding values for r.

Exercise: Consider the scatter plot below. What, approximately, is the sample correlation?



$r \approx 0.96$

$|r|$ close to 1 $\not\Rightarrow$ linear model is a good choice

The Parameter Estimates

⋆ **Exercise:** What route would you use to derive $\widehat{\beta}_0$ and $\widehat{\beta}_1$?

Using least squares to estimate β_0 and β_1 is definitely the standard approach, but it's not the only way to do it.

Choose $\hat{\beta}_0$, $\hat{\beta}_1$ to minimize $\sum_{i=1}^{n} \hat{\epsilon}_i^2$

If we were to follow through with this, we would find that the optimal estimates would result from minimizing the **residual sum of squares (RSS)**:

$$\mathrm{RSS}(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_i))^2 = \sum \hat{\epsilon}_i^2$$

And a little bit of calculus would show that

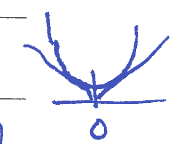
$$\widehat{\beta}_1 = \frac{\sum_i (y_i - \overline{y})(x_i - \overline{x})}{\sum_i (x_i - \overline{x})^2} = \frac{s_{xy}}{s_x^2} = \boxed{r_{xy}\left(\frac{s_y}{s_x}\right)}$$

and

$$\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}.$$

⋆ **Exercise:** Why do we use **least squares** to estimate the parameters in regression?

① Derivations are simple via Calculus, no need for numerical optimization



② Least squares gives the MLE (maximum likelihood estimator) under assumption ϵ_i are iid $N(0, \sigma^2)$

③ Gauss-Markov Theorem: LS estimators are the best, linear, unbiased estimators of β_0, β_1 under the three assumptions given above

"best": smallest variance

⋆ **Exercise:** Suppose that the roles of the predictor and response variables were interchanged in this regression model. What would be the effect on the slope of the regression line?

It's not just the reciprocal

original slope (y is resp) $= r_{xy}\left(\frac{s_y}{s_x}\right)$

new slope (y is predictor) $= r_{xy}\left(\frac{s_x}{s_y}\right)$

"There are two regression lines"

1 Figure 7 illustrates this idea. On the same scatter plot, two regression lines are shown: The "dashed-dotted" line is the regression line for predicting GOOG excess return from the S&P 500 excess return. The "dashed" line is the regression line for predicting S&P 500 excess return from GOOG excess return.

6 The solid line is the **SD Line.** This has a slope equal to s_y/s_x.

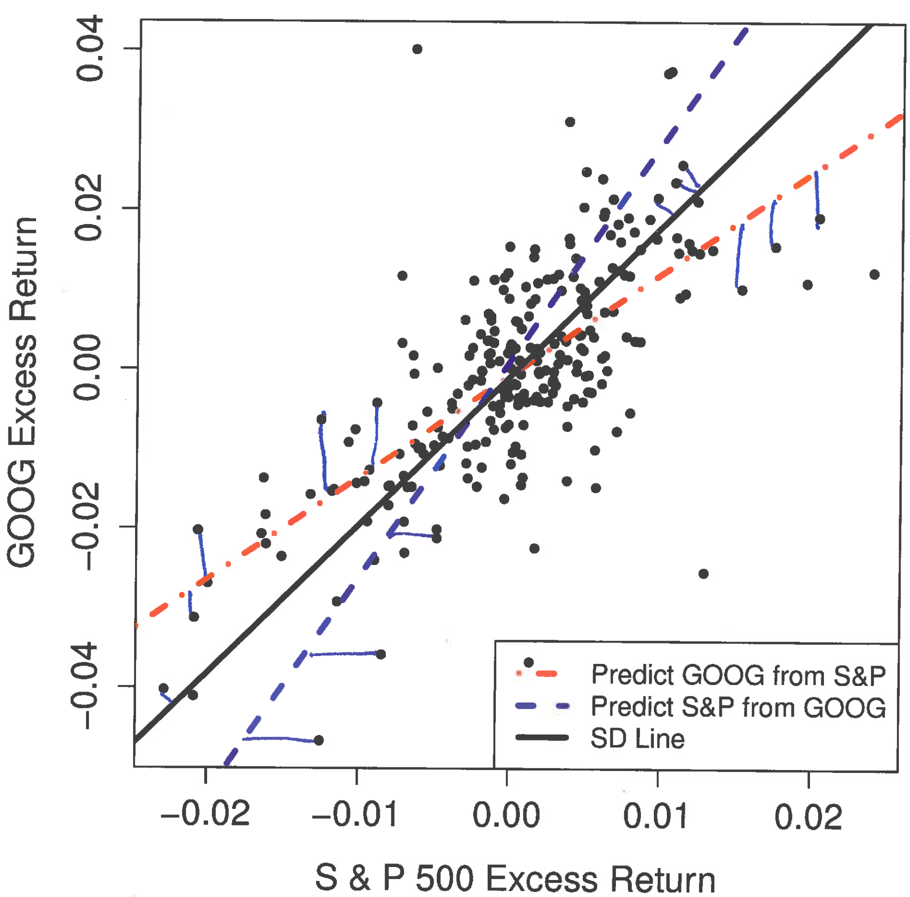


Figure 7: Daily excess returns for GOOG versus daily excess returns for S & P 500 for Jan. to Nov. 2012.

Estimating σ^2

when ϵ_i iid $N(0,\sigma^2)$

The MLE for σ^2 can be derived to be

$$\widehat{\sigma}^2_{\text{MLE}} = \left(\frac{1}{n}\right)\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 = \text{RSS}\Big/n.$$

This estimator is biased, but it can be adjusted to obtain an unbiased estimator

$$\widehat{\sigma}^2 = \left(\frac{n}{n-2}\right)\sigma^2_{\text{MLE}} = \left(\frac{1}{n-2}\right)\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 = \text{RSS}\Big/(n-2).$$

This is the estimator we will typically use, and what is reported by R and other software packages.

Exercise: Is there intuition as to why we divide by $(n-2)$?

$$\underset{\sim}{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \qquad \underset{\sim}{\widehat{y}_i} = \begin{bmatrix} \widehat{y}_i \\ \widehat{y}_2 \\ \vdots \\ \widehat{y}_n \end{bmatrix} \qquad \underset{\sim}{\widehat{\epsilon}_i} = \underset{\sim}{y_i} - \underset{\sim}{\widehat{y}_i}$$

$\underset{\sim}{y}$ lies in an n-dimensional space

$$\underset{\sim}{\widehat{y}} = \widehat{\beta}_0 \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + \widehat{\beta}_1 \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$\underset{\sim}{\widehat{y}}$ lies in a 2-dimensional space

$\underset{\sim}{\widehat{\epsilon}} = \underset{\sim}{y} - \underset{\sim}{\widehat{y}}$ lies in a n-2 dim. space, "$\underset{\sim}{\widehat{\epsilon}}$ has n-2 degrees of freedom"

There are two linear constraints on $\underset{\sim}{\widehat{\epsilon}}$

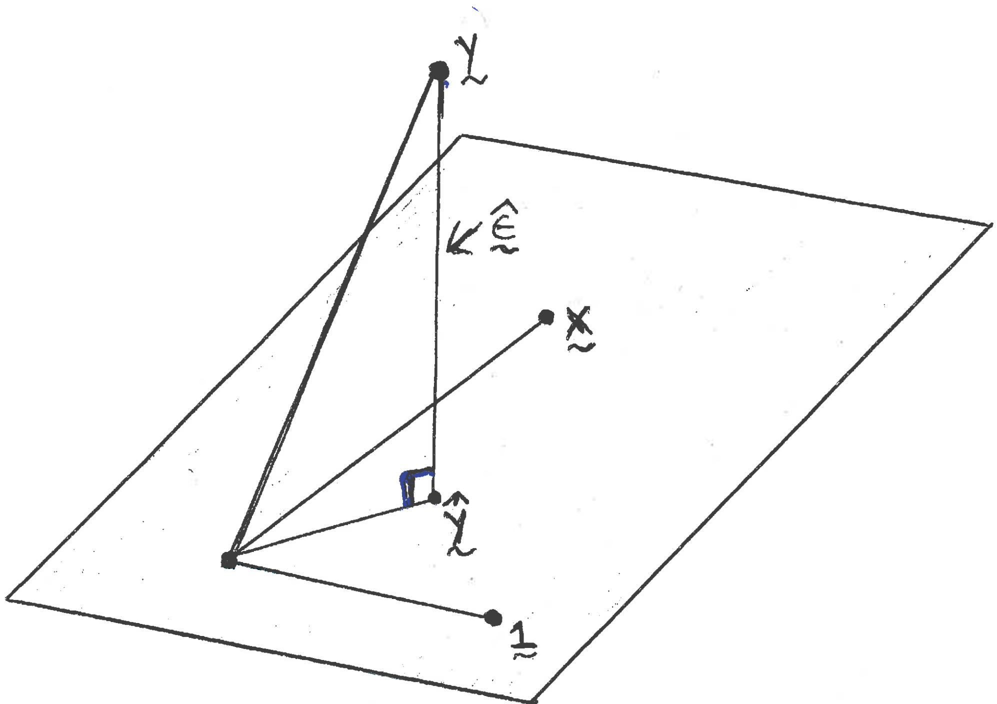


Figure 8: The geometric interpretation of least squares.

Figure 8 is a very useful geometric depiction of what happens when we do least squares. The vector of fitted values ŷ is the orthogonal projection of y onto the space spanned by the vectors 1 and x. (The rectangle represents all of the possible vectors that can be formed from linear combinations of 1 and x.)

⋆ **Exercise:** What is the sum of the residuals in OLS?

$\sum_{i=1}^{n} \hat{\epsilon}_i = \hat{\epsilon}^T \mathbf{1} = 0$, contrast that with $\sum_{i=1}^{n} \epsilon_i \neq 0$

but $E\left(\sum_{i=1}^{n} \epsilon_i\right) = 0$

under the assumptions

Inference Regarding β_1

Within the context of this simple model, the primary objective is to perform inference regarding β_1, since it captures information regarding the relationship (or lack thereof) between the response and the single predictor.

Here are some key results in this direction. Keep in mind that these are all true under the assumptions put forth for the simple linear regression model.

- $\widehat{\beta}_1$ is an unbiased estimator of β_1.
- The variance of $\widehat{\beta}_1$ is

$$V(\widehat{\beta}_1) = \frac{\sigma^2}{(n-1)s_x^2} \approx \frac{\widehat{\sigma}^2}{(n-1)s_x^2}. = \frac{\hat{\sigma}^2}{S_{xx}}$$

$S_{xx} = \sum (x_i - \bar{x})^2$

$S_x^2 = S_{xx}/(n-1)$

  This means that the standard error is

$$\mathrm{SE}(\widehat{\beta}_1) \approx \frac{\widehat{\sigma}}{s_x\sqrt{(n-1)}}.$$

- Suppose that the true value of the slope is β_{10}. Then

$$T = \frac{\widehat{\beta}_1 - \beta_{10}}{\mathrm{SE}(\widehat{\beta}_1)}$$

Assuming ϵ_i iid $N(0, \sigma^2)$

  has the t-distribution with $(n-2)$ degrees of freedom. Of course, when n is sufficiently large, T will be approximately standard normal.